

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I U. Udhayashree-192521448 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U.Udhayashree

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

ANSWER

(a) Purpose of Grey Level Transformation

Grey level transformation is used to improve the appearance of an image by modifying the intensity values of pixels. It helps in:

- Enhancing image contrast.
- Highlighting important features.
- Making dark or bright regions more visible.
- Preparing images for further processing.

(b) Negative Transformation

Formula:

$$s = 255 - r$$

Given:

$$r = [50, 100, 150, 200]$$

Calculations:

- $255 - 50 = 205$
- $255 - 100 = 155$
- $255 - 150 = 105$
- $255 - 200 = 55$

Answer:

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155
150	105
200	55

Output: [205, 155, 105, 55]

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log_{10}(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

ANSWER

(a) Purpose

Log transformation expands low-intensity (dark) pixel values while compressing high-intensity values.

This makes details in darker regions easier to see.

(b) Calculation

Formula:

$$s = c \log(1 + r)$$

Given:

- $c = 10$
- $r = [0, 10, 50, 100]$

Using natural logarithm (ln):

- $r = 0$

$$s = 10 \ln(1) = 10(0) = 0$$

- $r = 10$

$$s = 10 \ln(11) \approx 10(2.3979) = 23.98$$

- $r = 50$

$$s = 10 \ln(51) \approx 10(3.9318) = 39.32$$

- $r = 100$

$$s = 10 \ln(101) \approx 10(4.6151) = 46.15$$

r	s
0	0.00
10	23.98
50	39.32
100	46.15

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

ANSWER

(a) Purpose of Histogram Equalization

Histogram equalization improves image contrast by redistributing the intensity values so that the histogram becomes more uniform. It enhances details, especially in low-contrast images.

(b) Cumulative Distribution and Equalization

Given Histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels:

$$N = 2 + 2 + 4 + 8 = 16$$

For a 3-bit image:

$$L = 8$$

Formula:

$$s_k = (L - 1) \times CDF$$

Intensity	Frequency	Cumulative Frequency	CDF	Equalized Value
0	2	2	$2/16 = 0.125$	$7 \times 0.125 = 0.875 \approx 1$
1	2	4	$4/16 = 0.25$	$7 \times 0.25 = 1.75 \approx 2$
2	4	8	$8/16 = 0.50$	$7 \times 0.50 = 3.5 \approx 4$
3	8	16	$16/16 = 1.00$	$7 \times 1 = 7$

Equalized Intensities:

- $0 \rightarrow 1$
- $1 \rightarrow 2$
- $2 \rightarrow 4$
- $3 \rightarrow 7$

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

ANSWER

(a) Purpose

Spatial smoothing reduces image noise and small variations by replacing each pixel with the average of its neighboring pixels.

(b) 3×3 Averaging Filter

Matrix:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 50 & 60 & 70 \end{bmatrix}$$

Average Filter:

$$\frac{1}{9}$$

Sum of all values:

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Center pixel:

$$\frac{270}{9} = 30$$

Answer: 30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$, $\begin{bmatrix} -1 & 4 & -1 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$

to the matrix:

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

Compute the output at the center.

ANSWER

(a) Purpose

Sharpening enhances edges and fine details by emphasizing rapid changes in intensity. It makes images appear clearer and more focused.

(b) Laplacian Mask

Mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Calculation:

$$(4 \times 20) - (10 + 10 + 10 + 10) \\ 80 - 40 = 40$$

Answer: 40